LQR, iLQR, MPC... what should I choose??

01 – Motivation & Problem

Underactuated systems — like legged robots or lightweight manipulators — must stabilise dynamics through nonlinear coupling alone. The rotary double pendulum is the canonical minimal benchmark: two degrees of freedom, one actuator, an unstable upright equilibrium.

The question: which optimal control strategy actually wins under realistic conditions?

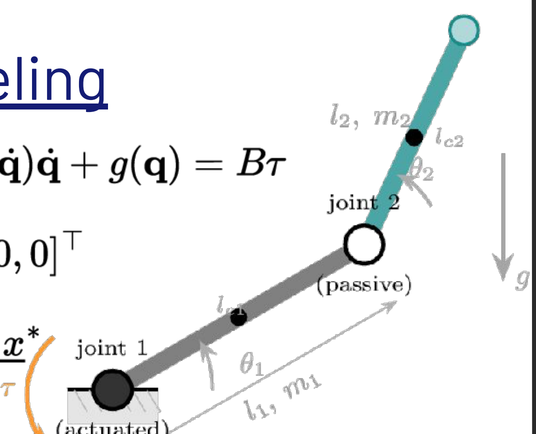
02 – Modeling

$$M(\mathbf{q})\ddot{\mathbf{q}} + C(\mathbf{q},\dot{\mathbf{q}})\dot{\mathbf{q}} + g(\mathbf{q}) = B\tau$$

$$x^* = [\pi/2, 0, 0, 0]^\top$$

Linearised about x^*

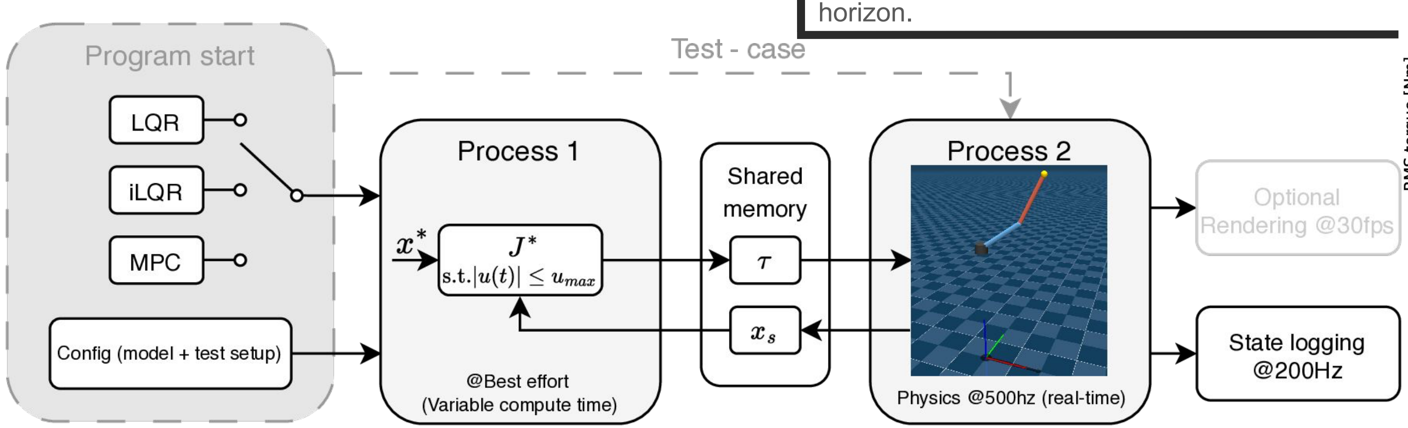
$$\dot{x} = Ax + B_u\tau$$



03 – Simulation & Evaluation

Test scenarios applied identically to all three controllers.

Perturbation	Levels
Sensor noise (σ)	0.01 / 0.05 / 0.10 rad
Torque limit ($u_{\max}$)	5 / 2 / 1 N·m
Mass mismatch	±10 / 20 / 30 %
Impulse	0.5 / 1.0 / 2.0 N·m·s
Combined stress	noise + torque + mismatch

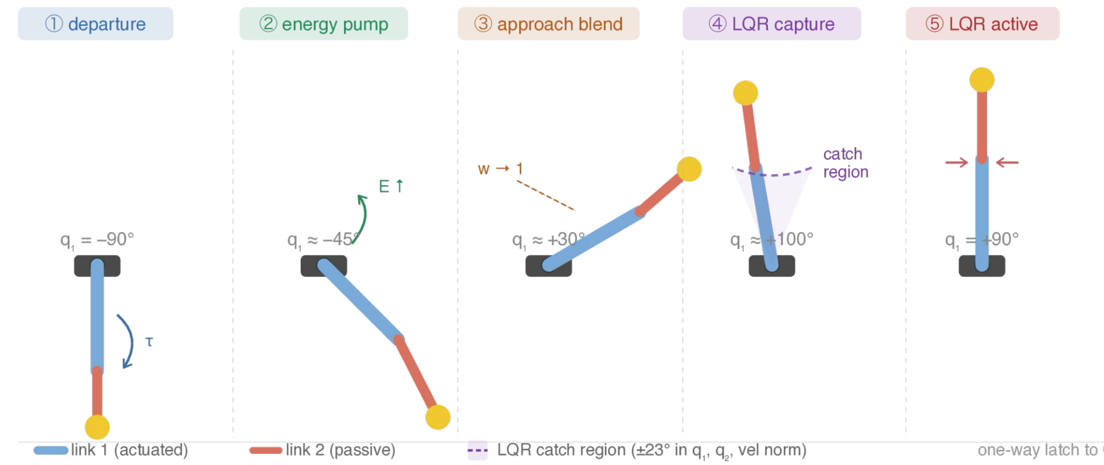


04 – LQR

Linear feedback at the upright equilibrium.

Operates on the *linearised* dynamics around x*. Gain K solved once from the algebraic Riccati equation. Wrapped in an energy-shaped swing-up that delivers the state into the linear region, where a four-condition catch latches LQR on.

Result: Five phases swing the pendulum from hanging to upright. Without the swing-up, pure LQR saturates the actuator and converges to the wrong equilibrium.

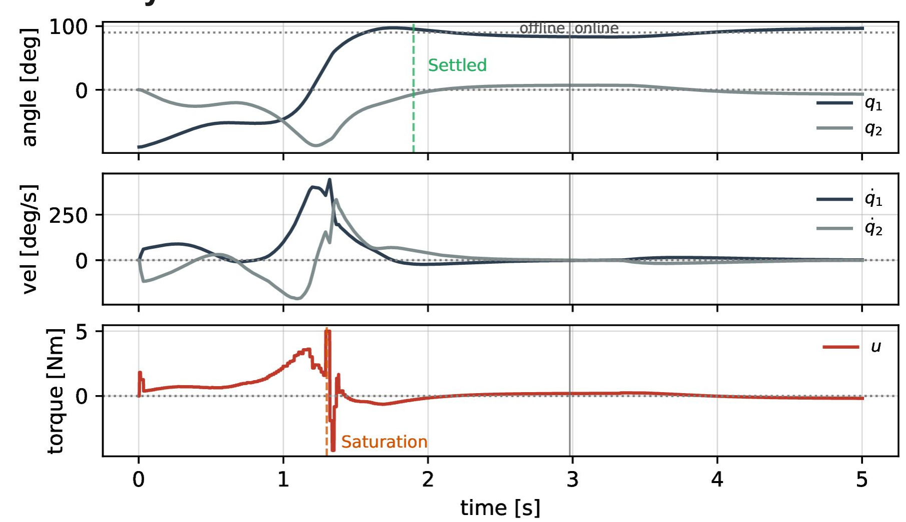


05 – iLQR

Trajectory optimisation on the full nonlinear dynamics.

Operates on the *full nonlinear* dynamics, re-linearised along the trajectory each iteration. Offline solve produces a nominal swing-up plus time-varying feedback gains; a short-horizon online phase replans during stabilisation.

Result: Offline plan reaches the upright in 1.5 s; the online phase holds it. Saturation episode at t ≈ 1.3 s when gravitational and Coriolis loading peak together.

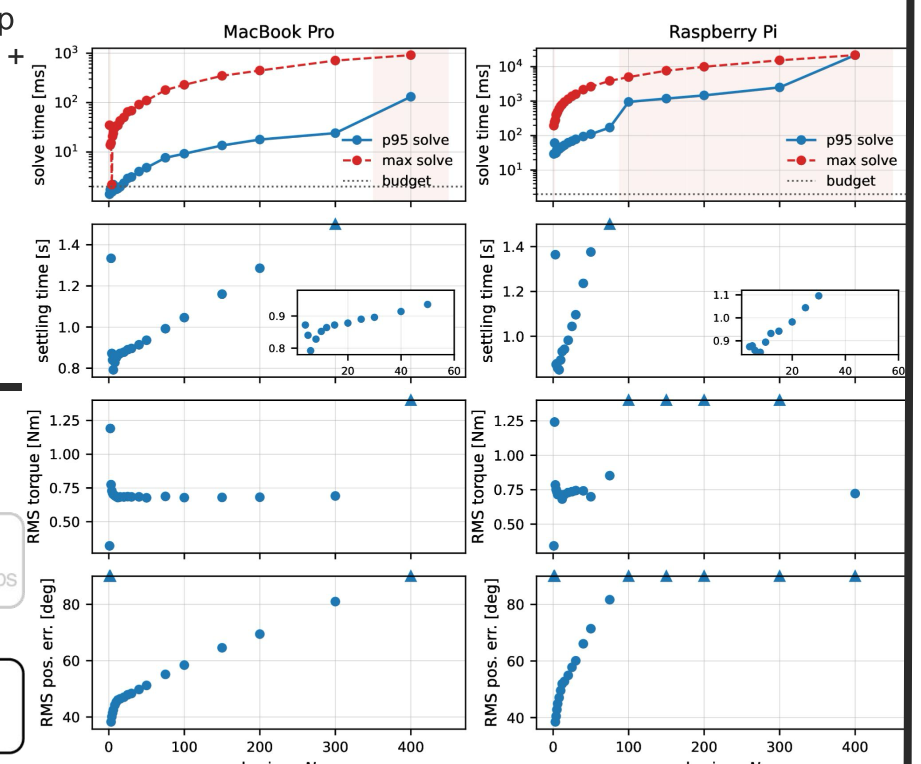


06 – MPC

Constrained optimisation at every control step.

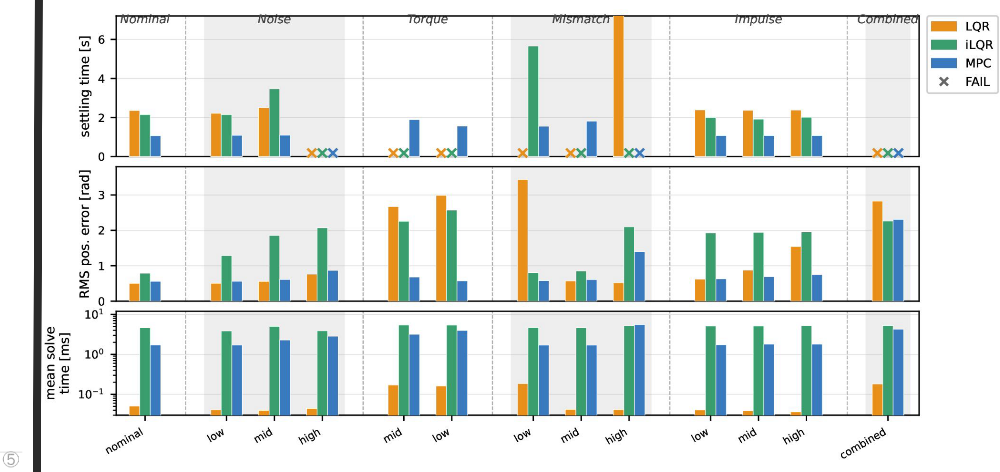
Solves a finite-horizon NLP at each step on the *full nonlinear* dynamics (CasADi + IPOPT, N = 20). DARE-based terminal cost provides the infinite-horizon cost-to-go; torque limits enter the optimisation as box constraints.

Result: Constraint awareness shows up as a sharp feasibility cliff at $u_{max} \approx 1$ N·m — below it, no swing-up trajectory exists within the horizon.

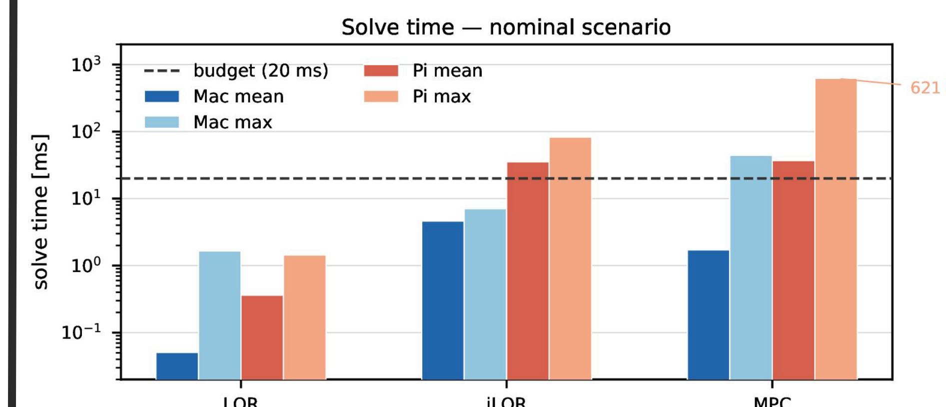


07 – Comparative Analysis

Stabilisation envelope: MPC succeeds in 10/13 scenarios by enforcing constraints inside the optimisation. LQR and iLQR each succeed in 7/13 — but for structurally different reasons.



Compute: LQR is trivially cheap on any hardware. iLQR's cost is bounded and predictable. MPC's worst-case latency is unbounded and explodes on the Pi.

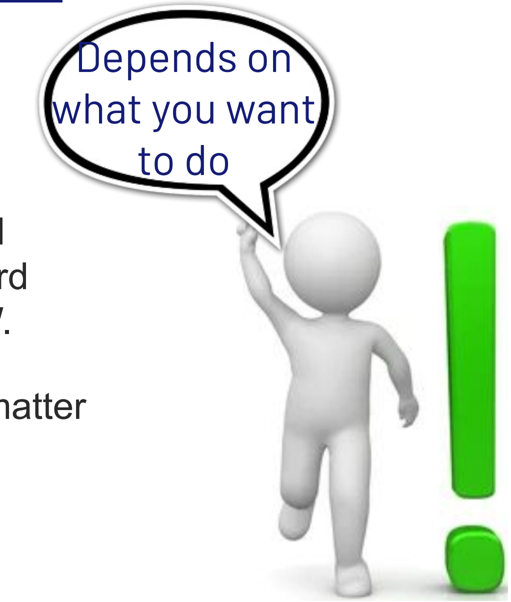


08 – So, which one?

Choose LQR if the system stays near equilibrium with low-noise sensing.

Choose iLQR if you need full nonlinear capability with hard real-time on embedded HW.

Choose MPC if constraints matter and you have compute headroom.



Sean Andersson, PhD